

Nikos Mavrapidis.

SOFTWARE ENGINEER · SYSTEMS & APPLIED AI

Software engineer with nearly three years of professional experience across three domains: embedded C/C++ on a carrier-grade Linux routing platform, Java and Python engineering on a large industrial printing suite, and satellite Earth Observation software for a European space-sector project—plus co-founder of Lope, an AI-native recruitment platform. Systems-level background in networking protocols, high availability, third-party integration, verification, and root-cause analysis. Core strength: understanding how components interact across protocol, operating-system, application, and third-party boundaries, and making them work together as one reliable system.

EXPERIENCE

Aug 2026—Present

Software Engineer · Space & Earth Observation · SoftCom International · European space-sector project

- Work on satellite data discovery systems built on Earth Observation metadata, STAC catalogues, and cloud-hosted imagery archives, including the retrieval architecture behind them.
- Built Earth Observation grounding through NASA ARSET training: sensing principles, spatial and spectral resolution, sensor and orbit trade-offs, and processing levels.

Oct 2025—Sep 2026

Co-founder · Lope · three-person founding team, alongside SoftCom

- Built the retrieval architecture end to end: hybrid semantic search on a self-hosted Milvus vector database with domain-adapted embeddings and rank fusion, grounded RAG with server-verified citations, and LLM-based document extraction.
- Owned the production architecture—Next.js, TypeScript, and PostgreSQL with row-level-security multi-tenancy, event-driven pipelines, and unit and end-to-end CI—and replaced a model-driven agent loop with a deterministic orchestrator that validates every output.
- Used in production by recruiting agencies; 98.7% enrichment success over ~3,000 profiles and 1,272 passing tests at handover.

Jun 2025—Jul 2026

Software Engineer · Industrial Printing · SoftCom International · industrial printing client · Java developer, then test automation engineer

- Delivered a backend and front-end module extension for a large distributed industrial printing suite; modernised legacy modules by reverse-engineering undocumented behaviour and resolved high-impact defects through peer-reviewed pull requests.
- Owned a critical memory-leak investigation end to end: VisualVM profiling, heap, log, and resource-trend evidence, analysis presented to the developers, fix validation, and automated regression coverage.
- Designed test-suite architecture and standards, built reusable keyword libraries—including custom Java keywords for Robot Framework—and added UI and REST automation across four product modules.
- Debugged failures across SUSE Linux Enterprise Server and Windows, managed the VMs and containerised environments used to validate releases, maintained Jenkins jobs and Gradle/Maven builds, and worked on site at the client's offices abroad.

Dec 2023—Jun 2025

Software Engineer · Embedded Linux & Networking · SoftCom International · telecom networking client

- Developed carrier-grade embedded Linux networking software in C/C++ across the full lifecycle on a high-availability IP/MPLS routing platform built with Yocto.
- Delivered Hitless Software Upgrades end to end with the HITSU team—design, implementation, integration, validation, deployment, and documentation—carrying live service state across versions and cutting upgrade interruption from hundreds of milliseconds to zero.
- Integrated and adapted a third-party carrier-grade routing stack, resolving interface and compatibility problems around software the team did not own.
- Validated control-plane and data-plane behaviour with Wireshark and Spirent TestCenter, root-causing issues across protocol, software, and platform boundaries; automated regression checks in Python.

SELECTED PROJECTS

Camelia Wellness · Client platform · Full-stack · 2025

A bilingual service and booking experience rebuilt around clearer journeys, discoverability, and operationally safe releases. Led UX and implementation across booking, SEO, responsive delivery, and versioned GCP/Cloudflare deployment with rollback.

Worthify · 1st Greek AI Hackathon · ACEin-AUEB · 2023–2024

A used-car valuation product trained on more than 100,000 listings and designed to make model output useful to everyday buyers. Co-created the platform across data, modelling, evaluation, and the product surface: validated at R^2 0.97 and roughly 6% median absolute percentage error, served through a REST API in a full-stack web app.

Adaptive Museum Experience · BSc thesis · Browser computer vision

A context-aware museum prototype that adapted content using browser-based visual interaction and was evaluated with 13 participants. Designed and built the experience end-to-end; 84.6% of participants reported feeling more knowledgeable after using it.

RETRIEVAL & AI R&D

Netflix Catalog Search · Lexical → semantic → hybrid → rerank

An offline catalog search that puts lexical, dense, hybrid, and two-stage retrieval side by side over the same corpus and measures each one. **nDCG@10 0.72 · Recall@10 0.78 · MRR 0.76**

News Evidence Retrieval · Passage retrieval & reproducible eval

Hybrid passage retrieval that surfaces supporting evidence for a claim over a news corpus — an evidence finder, deliberately not a fact-checker. **MRR 0.84 · Hit@1 0.78 · nDCG@5 0.63**

SKILLS

Applied AI & retrieval

Hybrid search & rank fusion, Vector databases (Milvus) & embeddings, RAG & LLM integration, Cross-encoder reranking, STAC & Earth Observation metadata

Test & verification

Robot Framework & Selenium, REST API & UI automation, VisualVM profiling & root-cause analysis, Playwright, Containerised test environments

Systems & networking

C/C++ · embedded Linux (Yocto), IP/MPLS, TCP/IP, control & data plane, Wireshark & Spirent TestCenter, Java, Python, Bash, SQL, TypeScript, Next.js & React

Delivery & data

Jenkins / GitLab CI, Gradle, Maven & Artifactory, Docker, PostgreSQL (row-level security), AWS & GCP

EDUCATION

2017–2022

BSc in Informatics & Telecommunications · National and Kapodistrian University of Athens

240 ECTS · Grade 7.67 / 10. Thesis: a web experience exploring ubiquitous computing, built for the department's Museum of IT & Telecommunications. [Read thesis](#)

CERTIFICATIONS & TRAINING

AWS Certified AI Practitioner · Amazon Web Services · 2026

Decoding DevOps: Linux, networking, containers, IaC, CI/CD, AWS & GCP (60+ hours) · Udemy · 2026

Cybersecurity Fundamentals (multi-course program) · Cyber Defense Architecture · IBM · 2026

Introduction to Cybersecurity · Networking Basics · Networking Devices and Initial Configuration · Cisco · 2025

Networking Basics and Security · IBM · 2025

MPLS L3VPN · Employer-funded · 2024

Software Engineering Basics for Everyone · IBM · edX · 2023

High-Dimensional Data Analysis · HarvardX · edX · 2023

Data Visualization · Coursera · 2021

COMPETITIONS & COMMUNITY

1st Greek AI Hackathon · Athens Center for Entrepreneurship and Innovation · AUEB · Feb 2024

Student volunteer, MobileHCI 2023 · ACM SIGCHI · Sep 2023

Google Hash Code 2022 · Athens hub hosted by NTUA · Feb 2022

Hackathon 2022 · Team participant · Feb 2022